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Material Confidence: Which Outcomes Can Tell Felt from Seen AI Confidence?

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ABSTRACT

AI assistants and conversational search systems increasingly attach a confidence to the information they return, and users must decide whether to accept it. When answers are spoken or the screen is out of view, confidence could be conveyed by touch instead, for example as a solid, grained or crumbly vibration. Before such displays are tested with people, one must know which outcome can tell them apart. We show how to settle this with existing data. Re-analysing a public dataset of 35,670 human–AI decisions, we find that acceptance of advice rose with its displayed confidence, yet on disagreements people ended with the correct answer less often than if they had always accepted it. Simulating a four-condition study (no cue, numeric, visual, vibration) from these decisions shows that the appropriate-reliance rate barely registers a confidence cue: in this design, even perfect use of the cue would raise it by only about three points. Assuming people decode the vibration on 80% of trials and act on it as on a visual cue, a test of reliance sensitivity—how steeply acceptance rises with the signalled confidence—detected that the vibration was used in 99% of simulated 96-person studies, and a test of the appropriate-reliance rate on the same studies in 7%. We report no tactile user data: the contribution is a way to choose the outcome, and the sample size, before such a study is run.

CCS CONCEPTS

• **Information systems → Users and interactive retrieval**; • **Human-centered computing → Empirical studies in HCI**; *Haptic devices.*

KEYWORDS

AI-assisted information access, confidence displays, appropriate reliance, evaluation methodology, simulation, power analysis, vibrotactile feedback

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1 INTRODUCTION

Generative AI now answers many information needs directly, and systems increasingly signal how confident they are in an answer [10, 16]. Whether a user accepts or rejects AI-provided information should depend on that signal: appropriate reliance means taking the answer when it is likely to be right and keeping one's own judgement otherwise [12, 13]. Confidence is almost always shown on a screen, as a percentage, a bar or a hedge in the text. Spoken conversational search has no screen to show it on [14], and on phones and wearables the screen is often out of view. Touch is another channel: small sets of vibration patterns can be learnt and identified without looking [4–6].

Whether people can use a felt confidence as they use a seen one is an empirical question, but testing it first requires an outcome that could reveal a difference. We show that the usual outcome—how often people end with the correct answer—is nearly blind to the display in a calibrated design, while an outcome that is rarely primary, how steeply acceptance rises with the signalled confidence, is not. The first measures a downstream benefit and the second what a cue acts on, so some gap is expected; what matters for study design is how large it is and why. Our contributions are (1) a re-analysis of public human–AI decisions showing how acceptance tracked displayed AI confidence; (2) a data-grounded user-simulation method that predicts, before a study, which outcomes can distinguish confidence displays and how many participants each needs, with a ledger of what is observed, fitted and assumed; and (3) its result for vibrotactile confidence, with a protocol ready for preregistration.

2 BACKGROUND

Confidence in AI-provided information. Displaying a model's confidence can help people calibrate their trust [16]. In LLM answers, first-person expressions of uncertainty reduced how often people agreed with the system and improved their accuracy [10], whereas confidently stated answers raise acceptance, including of wrong ones [7]. Reliance is usually scored on disagreements between a person's own answer and the AI's: switching to correct advice, keeping a correct answer against wrong advice, or both together [13]. Explanations and other additions often raise acceptance regardless of correctness [2].

Felt confidence. Tactile icons encode information in rhythm, roughness or intensity [4]. Identification depends on the dimension: three rhythms were identified on 93% of trials and three roughness levels on 80% with a fingertip actuator [5]; a smartwatch number code reached 96% in the lab and 54% while running [6]. A felt code is therefore likely to be read less reliably than a printed number.

Simulation before evaluation. User simulation is an established way to evaluate information access systems [1]. Simulation is also the standard way to plan sample sizes for mixed-effects designs [9, 11], and design analysis asks what a study of a given size can reliably show [8]. These approaches take the outcome as given. We use simulated users earlier, to choose the outcome of a user study.

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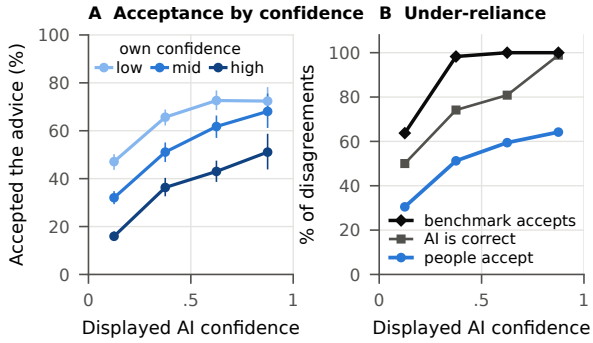


Figure 1: HAIID, 10,824 disagreements. (A) Share accepted by displayed AI confidence and the person's own confidence (95% participant-cluster bootstrap intervals). (B) How often the advice was correct, the confidence-only benchmark accepted it, and people accepted it.

3 STUDY 1: DISPLAYED AI CONFIDENCE AND ACCEPTANCE

Data and analysis. The Human–AI Interactions Dataset (HAIID) [15] records 35,670 decisions on five binary tasks. Participants answered on a slider whose sign gave the label and whose distance from the centre gave their confidence, then saw advice, attributed to an AI or a human, as a marker on the same scale—its distance from the centre displayed its confidence—and answered again. The advice was the average answer of about 50 earlier crowd workers, moved at random on half the trials by a fifth of the scale towards or away from the correct label, and participants were randomly assigned to see it labelled as AI or as human advice [15]. The tasks are judgements about images, texts and census records (such as an artwork's period or whether a sentence is sarcastic) rather than search tasks, but each decision has the structure that matters here: accepting or rejecting an answer offered with a stated confidence. We analysed the four crowd tasks (34,342 decisions by 1,088 people) and their 10,824 *disagreements*, on which the first answer and the advice pointed to different labels. A person *accepts* the advice if the final answer takes its side; the *appropriate-reliance rate* (ARR) is the share of disagreements that end with the correct source [13]. We fitted logistic mixed models with `lme4` [3] (participant and item random effects, participant slopes) and compared people with a cross-validated *confidence-only benchmark* that accepts whenever the displayed and own confidence predict the advice to be right.

Results. Acceptance rose with displayed confidence: each quarter of the scale was associated with 2.70 [2.47, 2.96] times the odds of accepting, also within items (2.48), and each quarter of the person's own confidence with 0.45 times the odds (Figure 1A). Displayed confidence was not varied independently of the item or of whether the advice was right, so these are associations, not causal effects. The source label was randomised: advice attributed to an AI was accepted more than identical advice attributed to a human (odds ratio 1.78 [1.41, 2.26]). Yet people accepted the advice on only 43.1% of disagreements although it was right on 64.8%. Their ARR, 55.3% [54.0, 56.5], was below that of always accepting (64.8%) and of the benchmark (65.9%), and under-reliance was largest where the advice was surest: in the top quarter of displayed confidence it was right on 98.9% of disagreements and accepted on 64.2% (Figure 1B). Self-reported trust in the advice tracked how often a person accepted it (Spearman $\rho = 0.50$ [0.45, 0.55], $n = 1{,}077$) but hardly whether their acceptance was appropriate ($\rho = 0.15$), so questionnaires are no substitute for behavioural outcomes here. Acceptance thus tracks displayed confidence, so a comparison of displays has something to detect; but the confidence display was worth only 0.5 points of ARR [-0.5, 1.5] even to the benchmark, because accepting was almost always the better choice.

Table 1: Provenance of the simulation's components.

Component	Source
Items, first answers	**Observed**: HAIID items and real participants' answers on them
Acceptance policy	**Fitted** to the 10,824 HAIID disagreements (odds ratio 2.55 per quarter of the confidence scale)
Same policy in every channel	**Assumed**: channel invariance (strength of the response to confidence varied ×0.5–1.5)
Cue decoding	**Assumed**: numeric .99, visual .97, vibration mean μ from 0.34 to 0.99 (main 0.80)
AI accuracy per level	**Design**: calibrated by construction

4 STUDY 2: CHOOSING THE OUTCOME BY SIMULATION

The simulated study. We simulate the study we plan to run. Each participant completes four 24-trial blocks, one per channel (no cue, numeric, visual bars, vibration), in a balanced Latin-square order, on 96 HAIID art, city and sarcasm items. The AI signals 63%, 75% or 88% and is calibrated by construction: within a block it is right on 5, 6 and 7 of the 8 trials at each level. The vibrations are three 450 ms patterns: one continuous pulse ('solid', 88%), three pulses ('grained', 75%) and seven short pulses ('crumbly', 63%).

What is observed, fitted and assumed. Table 1 lists the provenance of each component. The central assumption, which we call *channel invariance*, is that people act on the level they perceive in the same way in every channel, so a felt cue can lose influence only by being misread; whether that holds is what the planned study tests. Each simulated disagreement is scored under all four channels, so channels are compared on identical people and items. Outcomes are ARR and *reliance sensitivity*: the slope of acceptance across the three levels in a logistic mixed model with participant-specific slopes and own confidence as a covariate, compared with the same people's no-cue slope. An *ideal user*, who accepts whenever confidence alone makes the advice more likely right than wrong, gives the most any cue could add to ARR in this design. For power we simulated 200 complete studies per cell at 48–120 participants and at 288, redrawing the fitted parameters from their sampling distribution for each study, and analysed each with the planned models (one-sided $\alpha = .025$ for whether a cue is used; a 90% interval, one-sided $\alpha = .05$, for non-inferiority).

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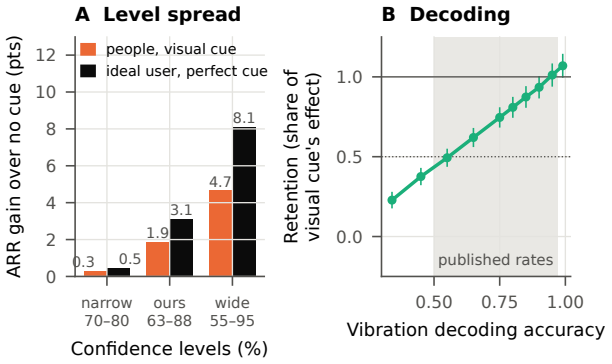


Figure 2: Two design levers (a separate simulated sample of 2,400 people, so values differ slightly from the main sample's). (A) ARR gain over no cue for three spreads of the confidence levels. (B) Share of the visual cue's gain in reliance sensitivity that a vibration retains, by mean decoding accuracy (±1 SE), under channel invariance; the band spans published identification rates for comparable tactile codes.

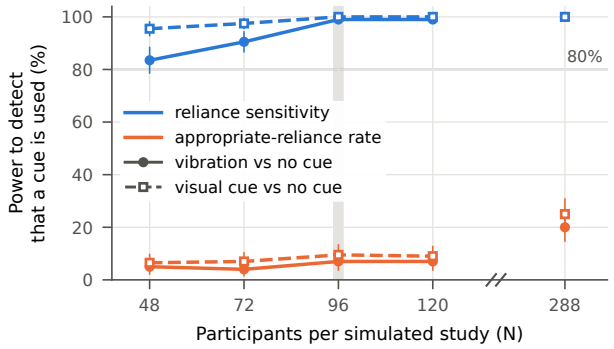


Figure 3: Power to detect that a cue changes reliance relative to no cue, on the same 200 simulated studies per sample size ($\mu = 0.80$; bars: Monte Carlo 95% intervals; grey band: planned N). The axis is broken between 120 and 288.

ARR barely registers the cue. Expected ARR was 55.0% with no cue, 57.0% with a visual cue and 56.5% with vibration. Even an ideal user reading the cue perfectly gains only about three points (Figure 2A), and a vibration that carries no information at all scores only 2.3 points below a visual cue. The ceiling is set by the design rather than by people: with confidence levels spread to 55/75/95%, the ideal user's gain rises to 8.1 points, and modelled people's to 4.7.

Reliance sensitivity registers it. Model slopes were 0.06 logits per level with no cue and 0.90 with a visual cue; vibration decoded at $\mu = 0.80$ retained 0.76 of the visual cue's gain over no cue. Under channel invariance, retention is set by decoding (Figure 2B): about one half at $\mu = 0.56$ and 0.9 at $\mu = 0.87$, so a felt code needs identification accuracy near the top of published rates [5, 6] to match bars. Across 9 alternative assumptions—a weaker or stronger response to confidence, another mapping of the levels onto HAIID's scale, no term for advice correctness, a vibration that adds or removes authority, other no-cue behaviour and less accurate visual decoding—retention stayed between 0.76 and 0.88.

A pitfall. Without any cue, acceptance still rose across the hidden levels (50.8% to 55.2%): where the AI is more often right, more disagreements are ones on which the person was wrong, less confident and more willing to switch. A raw comparison of acceptance across levels would mistake this for use of the cue; adjusting for own confidence and comparing each cue with the same people's no-cue slope removes it.

Power. Figure 3 answers one question—is a cue used?—with each outcome on the same simulated studies. With 96 participants, and under channel invariance, the sensitivity test detected that the vibration was used in 99% of studies (90% if decoding averages 0.70), the ARR test in 7%. An ARR test would need about 1,820 participants for 80% power, and about 1,200 even for a visual cue; telling an uninformative vibration from bars on ARR would need about 790. Showing that vibration is used *as well as* bars (non-inferiority of the slope, margin 0.40 logits, about half the visual cue's gain over no cue) is harder: 37% power at 96 and 81% at 288. These power figures are predictions conditional on the assumed parts of Table 1, which Study 3 can falsify; the ideal-user ceiling and Study 1 rest on none of them.

5 DISCUSSION

Evaluating confidence displays for information use. In calibrated designs with moderately spaced confidence levels, the appropriate-reliance rate is close to blind to how confidence is displayed: a study of typical size would be unlikely to distinguish a working display from a useless one, and a null result on ARR says little about whether people used the cue. That an outcome measuring the response to confidence detects a cue more easily than one measuring its benefit is expected; what the analysis adds is the size of the gap in a typical design and its cause, a ceiling of about three points. Three practices follow for studies of confidence in AI-provided information, whatever the modality. Make reliance sensitivity, measured against a no-cue baseline, the primary outcome when the question is whether a display is used; it shows use, not benefit. Choose confidence levels that straddle people's own accuracy, so that confidence changes the best action often and ARR has room to show whether a display helps. And compute the ideal-user ceiling before the study: it depends only on the items, people's first answers and the levels, not on assumptions about how a display is used, and it shows at once whether ARR can separate displays at all. The same steps apply to verbal hedges, colour or sound.

Where felt confidence could matter. Our design puts every channel in vision's most favourable condition: a seated person looking at the screen. Felt confidence is aimed at information access without a screen or with the eyes elsewhere—spoken conversational search [14], answers on a watch or while walking [6]—where the alternative to a vibration is often no confidence signal at all. A study there needs the same outcome choice, and the ceiling and power calculations above can be rerun for its tasks.

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Planned study. Study 3 follows this analysis: a within-participant comparison of the four channels with 96 participants, using their own Android phones (web vibration) or a wrist band, with reliance sensitivity relative to no cue as the confirmatory outcome and the vibration–visual comparison as an estimation target. Its instrument and analysis code are complete: 52 scripted test sessions ran the study in a real browser (4,944 trials) with no deviations from the scheduled design, and the analysis pipeline excluded exactly the sessions built to fail. Before registration, a device test will check that phones render the patterns as specified (the seven-pulse pattern is the most likely to blur on a phone motor), and a short decoding pilot will estimate how well people tell them apart; the patterns will be revised if mean accuracy is below 0.70.

Limitations. The predictions rest on channel invariance, which the planned study tests rather than presumes. The AI is calibrated by construction, so the design cannot show whether felt confidence helps when the AI's confidence is wrong; there, a steeper reliance slope could mean worse decisions, and a miscalibrated condition would be needed. HAIID displayed confidence as a marker on the participant's own scale, its advice came from crowd averages, and its tasks were low-stakes; Study 3 reuses its items, so it tests the decision to accept an answer offered with a confidence, not information access itself. Decoding accuracies are borrowed from tactile studies with other devices. No person has yet felt our patterns, and neither the phone patterns nor the band have been measured on hardware, so we make no claim that felt confidence helps or harms reliance.

6 CONCLUSION

Before testing whether people can use an AI's confidence when they feel it, we asked what such a test would have to measure. With real decisions from a public dataset and a transparent simulation, the answer for calibrated designs with moderately spaced confidence levels is that the appropriate-reliance rate is unlikely to tell confidence displays apart at feasible sample sizes, while reliance sensitivity against a no-cue baseline could show with about a hundred participants whether a cue is used, provided people decode it reasonably well. Showing that a cue also improves decisions needs a design whose ceiling leaves room for it.

Materials and use of AI. Code, data, the study instrument and the protocol are provided as anonymised supplementary material and will be released publicly; HAIID is used under its MIT licence. Following the ACM policy on AI, we disclose that Claude (Anthropic) wrote and ran the analysis, simulation and instrument code under the authors' direction and drafted the text. All analyses were run and checked by script, every result in the paper is taken from the analysis outputs, and the authors take full responsibility for the content.

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